Barycentric interpolation and Gaussian quadrature

Homework 2

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1. Interpolation with the barycentric formula

Polynomial interpolation constructs a polynomial $p(x_i)$ that matches a given function at selected nodes. The interpolating polynomial p_n is the unique polynomial of degree at most n that satisfies

$$p_n(x_k) = y_k, \tag{1}$$

for nodes $x_0, ..., x_k$ and sampled values $y_k = f(x_k)$. Interpolation in the standard basis uses the Vandermonde matrix. However, this can be computationally expensive and numerically unstable, especially for high-degree polynomials or poorly chosen nodes. Interpolation costs approximately $\frac{2}{3}n^3 O(n^2)$ operations.

To solve these issues, we move from the standard basis to the more stable Lagrange basis. The Lagrange form describes the same polynomial directly in terms of the sampled valyes. For each node x_k, we define a Lagrange basis polynomial as

$$l_k(x) = \prod_{j=0, j \neq k}^{n} \frac{x - x_j}{x_k - x_j}. \tag{2}$$

It satisfies:

$$l_k(x_j) = \begin{cases} 1 \text{ if } j = k \ , \\ 0 \ \text{ else.} \end{cases} \tag{3}$$

The interpolating polynomial for $j = k$ is then:

$$p_n(x) = \sum_{k=0}^{n} y_k l_k(x). \tag{4}$$

For each node x_i, all terms except the i-th vanish:

$$p_n(x_i) = \sum_{k=0}^{n} y_k l_k(x_i) = y_i. \tag{5}$$

Polynomial interpolation constructs a polynomial that matches a given function at selected nodes. The Lagrange form provides an explicit expression for this polynomial, but evaluating it directly repeats many calculations. The barycentric form reorganizes the same polynomial so that reusable weights can be computed in advance, making repeated evaluation more efficient.

The cost of evaluating one $l_k(x)$ consists of multiplying approximately n factors. There are $n+1$ basis polynomials, so directly evaluating the Lagrange form costs $O(n^2)$ operations per argument.

The barycentric form avoids explicitly calculating every Lagrange basis polynomial $l_k(x)$. We use the following formula for barycentric Lagrange interpolation:

$$l(x) = \begin{cases} \frac{\sum\left(\frac{f(x_j)\lambda_j}{x} - x_j\right)}{\sum\left(\frac{\lambda_j}{x - x_j}\right)} \text{ if } x \neq x_j \ , \\ f(x_j) \text{ else.} \end{cases} \quad (6)$$

1.1. Chebyshev nodes

Polynomial interpolation using Chebyshev nodes minimizes the effect of Runge's phenomenon. Runge's phenomenon refers to oscillations near the edges of an interval that can occur when high-degree polynomial interpolation uses a set of equidistant interpolation points.

In our case, we interpolate using Chebyshev nodes generated on the interval $[-1, 1]$ with the formula

$$x_k = \cos\left(\frac{2k-1}{2n}\pi\right), \quad k = 0, 1, ..., n-1. \quad (7)$$

We also need the Chebyshev weights λ_j for barycentric interpolation. The weights are given by the following formula:

$$\lambda_k = (-1)^k \begin{cases} 1 \text{ if } 0 < 1 < n \ , \\ \frac{1}{2} \text{ if } i = 0 \ , \\ n \text{ else.} \end{cases} \quad (8)$$

We map the interval from $t \in [-1, 1]$ to $x \in [a, b]$ using an affine transformation[1]:

$$x = \frac{a+b}{2} + \frac{b-a}{2}t. \quad (9)$$

NOTE: There seems to have been a mismatch between the formulas for the Chebyshev weights and nodes. The provided node formula uses Chebyshev points of the first kind, whereas the provided weight formula uses Chebyshev weights of the second kind. We therefore use formulas of the second kind for both:

$$x_k = \cos\left(\frac{k}{n}\pi\right), \quad k = 0, 1, ..., n.$$
$$\lambda_k = (-1)^k \begin{cases} \frac{1}{2} \text{ if } k = 0 \ , \ n, \\ 1 \text{ else.} \end{cases} \quad (10)$$

[1] https://en.wikipedia.org/wiki/Chebyshev_nodes#Definition

1.2. Interpolation of the example functions

The goal was to determine a polynomial degree for which the maximum interpolation error is below 10^{-6}. For the first two functions, we tested consecutive degrees and found the first one that met the target. The third function required a much larger degree, so we estimated it from the observed convergence rate and verified the result at $n = 1.2 \cdot 10^6$. The results are shown in Table 1.

$f(x)$	interval	n	Error at $n-1$	Error at n
e^{-x^2}	$[-1, 1]$	10	$2.0811 \cdot 10^{-5}$	$8.2919 \cdot 10^{-7}$
$\frac{\sin(x)}{x}$	$[0, 10]$	13	$4.4543 \cdot 10^{-6}$	$1.3770 \cdot 10^{-7}$
$\lvert x^2 - 2x \rvert$	$[1, 3]$	1200000	$1.1938 \cdot 10^{-6}$	$9.9487 \cdot 10^{-7}$

Table 1: Degrees used to reach an error below 10^{-6}. For the third function, the earlier error was measured at $n = 10^6$.

Figure 1 shows the absolute interpolation error at the degrees listed in Table 1.

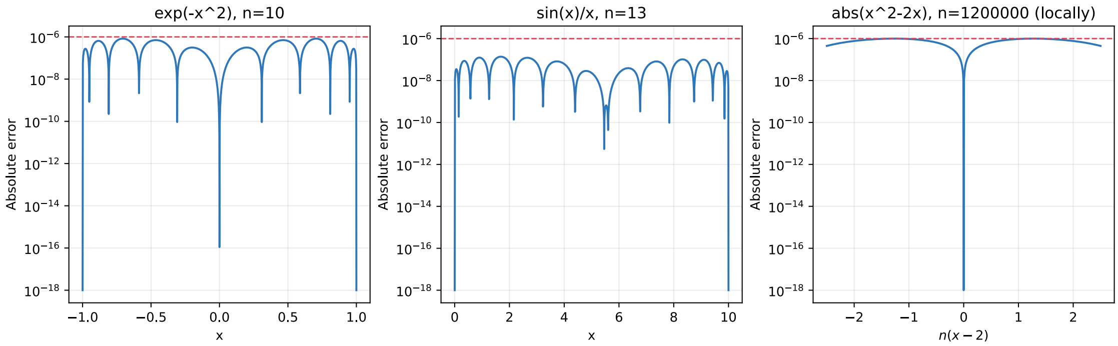

Figure 1: The curve shows the absolute interpolation error, and the dotted line marks the target of 10^{-6}.

The functions and their interpolants are compared in Figure 2.

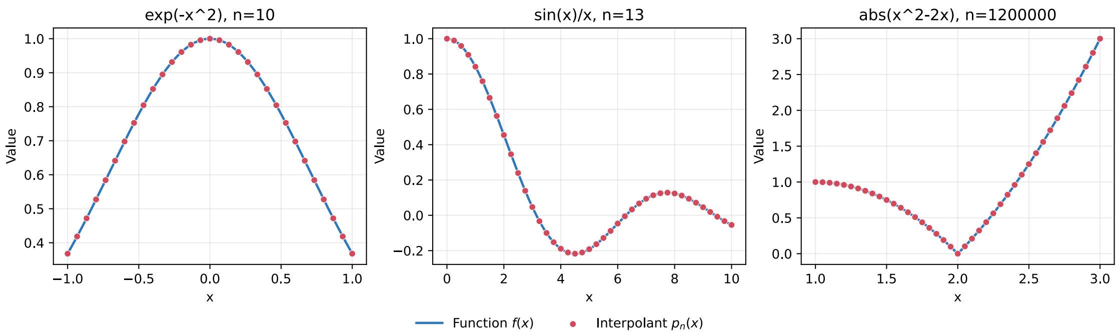

Figure 2: The blue curves show the original functions, while the red points are values of the corresponding interpolants.

For the first two smooth functions, the error decreases quickly as the polynomial degree increases because both functions can be approximated well by polynomials. However, the function $g(x) = |x^2 - 2x|$ has a corner at $x = 2$, where its first derivative is discontinuous:

$$g'_{r(2^-)} = -2 \quad g'_{r(2^+)} = 2. \tag{11}$$

I.e., the slope jumps from -2 to 2, which produces a corner. This matters because polynomials are smooth and have a continuous derivative everywhere. To reproduce the corner, the polynomial starts bending rapidly near $x = 2$, resulting in a large increase in the required degree.

In the experiments, the maximum error here decreases approximately as

$$E_n \approx \frac{C}{n}. \tag{12}$$

The experiments give $C \approx 1.194$[2]. To satisfy the required error $E_n < 10^{-6}$, we therefore need approximately

$$n > \frac{C}{10^{-6}} \approx 1.194 \cdot 10^6 \approx 1.2 \text{ million}. \tag{13}$$

This explains the large degree required for the third function. In Figure 3, the measured errors approximately follow a line proportional to $\frac{1}{n}$, confirming the much slower convergence.

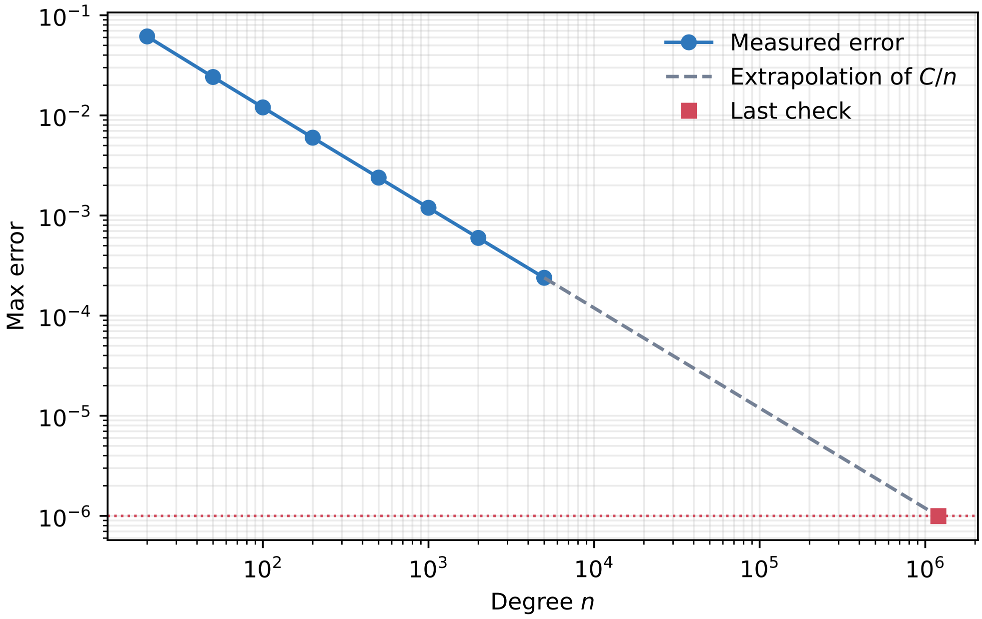

Figure 3: Convergence of the error as the polynomial degree increases.

[2]This can be seen in the convergence results found in `./results/hw2_part1.json`.

2. Gauss–Legendre quadrature

An integral can be approximated by a weighted sum

$$\int_a^b f(x)dx \approx \sum_{k=1}^{r} w_k f(x_k), \tag{14}$$

where x_k are the nodes and w_k are the weights.

Unlike the trapezoidal rule and Simpson's rule, Gauss–Legendre quadrature also determines the positions of the nodes. This allows it to integrate higher-degree polynomials exactly with the same number of function evaluations.

We derive the two-point Gauss–Legendre quadrature rule

$$\int_{-1}^{1} f(x)dx = Af(x_1) + Bf(x_2) + R_f, \tag{15}$$

including its error term R_f.

We consider polynomials and choose w_i and x_i so that the formula is exact for polynomials of the highest possible degree.

2.1. Deriving the two-point formula on $[-1, 1]$

Definition 2.1.1: The two-point formula for Gauss–Legendre quadrature on $[-1, 1]$ is

$$\int_{-1}^{1} f(x)dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) + R_f. \tag{16}$$

The error is

$$R_f = \frac{1}{135} f^4(\xi), \quad \xi \in (-1, 1). \tag{17}$$

Proof of Definition 2.1.1: We start with the two-point rule:

$$\int_{-1}^{1} f(x)dx \approx w_1 f(x_1) + w_2 f(x_2). \tag{18}$$

There are four unknowns: w_1, w_2, x_1, x_2. We substitute successive polynomials to find the solution.

For degree 0, we have $f(x) = 1$:

$$\int_{-1}^{1} 1 \,\mathrm{d}x = 2 = w_1 + w_2. \tag{19}$$

Hence $w_1 + w_2 = 2$.

For degree 1, we have $f(x) = x$:

$$\int_{-1}^{1} x \,\mathrm{d}x = 0 = w_1 x_1 + w_2 x_2. \tag{20}$$

Hence $w_1 x_1 + w_2 x_2 = 0$.

For degree 2, we have $f(x) = x^2$:

$$\int_{-1}^{1} x^2 \,\mathrm{d}x = \frac{2}{3} = w_1 x_1^2 + w_2 x_2^2. \tag{21}$$

Hence

$$w_1 x_1^2 + w_2 x_2^2 = \frac{2}{3}. \tag{22}$$

For degree 3, we have $f(x) = x^3$:

$$\int_{-1}^{1} x^3 \, \mathrm{d}x = 0 = w_1 x_1^3 + w_2 x_2^3. \tag{23}$$

Hence $w_1 x_1^3 + w_2 x_2^3 = 0$.

The full system of equations is:

$$\begin{aligned} w_1 + w_2 &= 2, \\ w_1 x_1 + w_2 x_2 &= 0, \\ w_1 x_1^2 + w_2 x_2^2 &= \frac{2}{3}, \\ w_1 x_1^3 + w_2 x_2^3 &= 0. \end{aligned} \tag{24}$$

We can now solve the system. Since the interval $[-1, 1]$ is symmetric, we seek a symmetric rule. Therefore, $x_2 = -x_1$ and $w_1 = w_2$.

We now use the x^2 equation:

$$x_1^2 + x_2^2 = \frac{2}{3}. \tag{25}$$

Substituting $x_2 = -x_1$ gives

$$\begin{aligned} 2x_1^2 &= \frac{2}{3} \\ x_1^2 &= \frac{1}{3}. \end{aligned} \tag{26}$$

Thus we can choose

$$x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}. \tag{27}$$

The complete two-point formula is

$$\int_{-1}^{1} f(x) \, \mathrm{d}x \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) + R_f. \tag{28}$$

The error is calculated as

$$R_f = \int_{-1}^{1} f(x) \, \mathrm{d}x - \left(f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)\right). \tag{29}$$

We already know that Gauss–Legendre quadrature is exact for every polynomial of degree ≤ 3. Therefore, the lowest-order derivative that can contribute to the error is the fourth derivative.

The first polynomial for which an error can appear is $f(x) = x^4$. Substituting it as before gives

$$\int_{-1}^{1} x^4 \, \mathrm{d}x = \frac{2}{5} \neq \frac{2}{9} = \left(-\frac{1}{\sqrt{3}}\right)^4 + \left(\frac{1}{\sqrt{3}}\right)^4, \tag{30}$$

so, unlike the cases for degrees 0 to 3, the two values are not equal.

Their difference is the error:

$$R_{x^4} = \frac{2}{5} - \frac{2}{9} = \frac{8}{45}. \tag{31}$$

This is only the error for degree 4, but we need a general error formula for an arbitrary function f. The general error theorem has the form

$$R_f = C f^{(4)}(\xi), \quad \text{for some } \xi \in (-1, 1). \tag{32}$$

We can determine C using $f(x) = x^4$. We already know that $R_{x^4} = \frac{8}{45}$ and $f^{(4)}(x) = 24$. Therefore

$$\begin{aligned} \frac{8}{45} &= C \cdot 24 \\ C &= \frac{1}{135}. \end{aligned} \tag{33}$$

Finally, we obtain the general formula

$$R_f = \frac{1}{135} f^{(4)}(\xi). \tag{34}$$

■

2.2. Deriving the two-point formula on $[a, b]$

We can go directly from $[-1, 1]$ to an arbitrary interval $[a, b]$ using the linear substitution

$$x(t) = \frac{b-a}{2} t + \frac{a+b}{2}, \quad t(x) = 2\frac{x-a}{b-a}. \tag{35}$$

The integral on $[a, b]$ can be expressed in terms of an integral on $[-1, 1]$:

$$\int_a^b f(x)\,\mathrm{d}x = \int_{-1}^1 f(x(t)) x'(t)\,\mathrm{d}t = \frac{1}{2}(b-a) \int_{-1}^1 f(x(t))\,\mathrm{d}t, \tag{36}$$

where $x'(t) = \frac{1}{2}(b-a)$.

Therefore, the Gauss–Legendre formula on $[a, b]$ is

$$\int_a^b f(x)\,\mathrm{d}x = \frac{1}{2}(b-a) \int_{-1}^1 f(x(t))\,\mathrm{d}t \approx \frac{b-a}{2} \left(f\left(x\left(-\frac{1}{\sqrt{3}} \right) \right) + f\left(x\left(\frac{1}{\sqrt{3}} \right) \right) \right). \tag{37}$$

Applying the two-point rule on $[-1, 1]$, where $t_1 = -\frac{1}{\sqrt{3}}, \quad t_2 = \frac{1}{\sqrt{3}}$, gives

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{2} \left(f\left(\frac{a+b}{2} - \frac{b-a}{2\sqrt{3}} \right) + f\left(\frac{a+b}{2} + \frac{b-a}{2\sqrt{3}} \right) \right). \tag{38}$$

Again, we determine the error formula R_f by testing a polynomial of degree 4. We already have the rule

$$\frac{b-a}{2} \left(f\left(m - \frac{b-a}{2\sqrt{3}} \right) + f\left(m + \frac{b-a}{2\sqrt{3}} \right) \right), \tag{39}$$

where $m = \frac{a+b}{2}$.

In this case, we choose the polynomial $f(x) = (x-m)^4$, because the interval is symmetric around m.

Now let

$$d = \frac{b-a}{2}, \quad a = m - d, \quad b = m + d. \tag{40}$$

The interval therefore runs from $m - d$ to $m + d$. We introduce the substitution $y = x - m$. The integration limits then become $a \to y = (m-d) - m = -d$ and $b \to y = (m+d) - m = d$.

The exact integral is

$$\int_a^b (x-m)^4\,\mathrm{d}x = \int_{-d}^d y^4\,\mathrm{d}y = \left[\frac{y^5}{5} \right]_{-d}^d = \frac{2d^5}{5}. \tag{41}$$

We now apply the Gauss–Legendre rule. The two nodes are $m \pm \frac{d}{\sqrt{3}}$, so

$$d\left(\left(-\frac{d}{\sqrt{3}}\right)^4 + \left(\frac{d}{\sqrt{3}}\right)^4\right) = d\left(\frac{d^4}{9} + \frac{d^4}{9}\right) = \frac{2d^5}{9}. \tag{42}$$

Therefore, the error for this fourth-degree polynomial is

$$R_f = \frac{2d^5}{5} - \frac{2d^5}{9} = \frac{8d^5}{45}. \tag{43}$$

Using the general error form $R_f = Cf^{(4)}(\xi)$ and the fourth derivative $f^{(4)}(x) = 24$, we obtain

$$\begin{aligned} \frac{8d^5}{45} &= C \cdot 24, \\ C = \frac{d^5}{135} &= \frac{1}{135} \cdot \left(\frac{b-a}{2}\right)^5 = \frac{(b-a)^5}{4320}. \end{aligned} \tag{44}$$

Thus

$$R_f = \frac{(b-a)^5}{4320} f^4(\xi), \quad \xi \in (a,b). \tag{45}$$

2.3. Deriving the two-point formula on $[0, h]$

This is a special case of $[a, b]$, where

$$a = 0, \quad b = h \tag{46}$$

.

In

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{2}\left(f\left(\frac{a+b}{2} - \frac{b-a}{2\sqrt{3}}\right) + f\left(\frac{a+b}{2} + \frac{b-a}{2\sqrt{3}}\right)\right), \tag{47}$$

we set $a = 0$ and $b = h$.

Then

$$\frac{a+b}{2} = \frac{h}{2}, \quad \frac{b-a}{2} = \frac{h}{2}. \tag{48}$$

Therefore,

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{h}{2}\left(f\left(\frac{h}{2} - \frac{h}{2\sqrt{3}}\right) + f\left(\frac{h}{2} + \frac{h}{2\sqrt{3}}\right)\right). \tag{49}$$

This is equivalent to

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{h}{2}\left(f\left(\frac{h}{2}\left(1 - \frac{1}{\sqrt{3}}\right)\right) + f\left(\frac{h}{2}\left(1 + \frac{1}{\sqrt{3}}\right)\right)\right). \tag{50}$$

The error follows in the same way. Substituting $a = 0$ and $b = h$ into

$$R_f = \frac{(b-a)^5}{4320} f^{(4)}(\xi) \tag{51}$$

gives

$$R_f = \frac{h^5}{4320} f^{(4)}(\xi), \quad \xi \in (0,h). \tag{52}$$

2.4. Composite two-point formula

We now construct the composite rule by dividing the interval into smaller parts.

Divide $[a, b]$ into n equal subintervals and let

$$h = \frac{b-a}{n}, \quad x_i = a + ih. \tag{53}$$

On each subinterval $[x_i, x_{i+1}]$, apply the two-point Gauss–Legendre rule. Since the midpoint of the subinterval is $\frac{x_i + x_{i+1}}{2} = x_i + \frac{h}{2}$, we get

$$\int_a^b f(x)\,\mathrm{d}x \approx Q_n = \frac{h}{2} \sum_{i=0}^{n-1} \left(f\left(x_i + \frac{h}{2} - \frac{h}{2\sqrt{3}}\right) + f\left(x_i + \frac{h}{2} + \frac{h}{2\sqrt{3}}\right) \right). \tag{54}$$

Thus, we split the integral into n subintervals, apply the Gauss–Legendre approximation on each one, and sum the results.

Hence

$$\int_a^b f(x)\,\mathrm{d}x = Q_n + R_n. \tag{55}$$

For one subinterval, the error is

$$R_i = \frac{h^5}{4320} f^4(\xi_i), \tag{56}$$

for some $\xi_i \in (x_i, x_{i+1})$.

Adding the errors from all subintervals gives

$$R_n = \frac{h^5}{4320} \sum_{i=0}^{n-1} f^4(\xi_i). \tag{57}$$

If f^4 is continuous, then for some $\xi \in (a, b)$ we can write $\sum_{i=0}^{n-1} f^4(\xi_i) = n f^4(\xi)$.

Substituting this into the formula for all subintervals gives

$$R_n = n \frac{h^5}{4320} f^4(\xi). \tag{58}$$

Using $nh = b - a$ and the definition of h, we can rewrite this as

$$R_n = \frac{(b-a)^5}{4320 n^4} f^4(\xi). \tag{59}$$

Therefore, the composite two-point Gauss–Legendre rule is fourth order:

$$R_n = O(h^4) = O(n^{-4}). \tag{60}$$

2.4.1. Error estimation by doubling the number of subintervals

Since the composite rule is fourth order, its error behaves approximately as

$$R_n \approx C h^4. \tag{61}$$

If we double the number of subintervals from n to $2n$, the step size is halved: $h \to \frac{h}{2}$.

Therefore,

$$R_{2n} \approx C \left(\frac{h}{2}\right)^4 = \frac{1}{16} C h^4 \approx \frac{1}{16} R_n. \tag{62}$$

Let I denote the exact value of the integral. Since $Q_n = I - R_n$ and $Q_{2n} = I - R_{2n}$, we can subtract the formulas:

$$Q_{2n} - Q_n = (I - R_{2n}) - (I - R_n) = R_n - R_{2n}. \tag{63}$$

Using $R_n \approx 16R_{2n}$, we obtain

$$Q_{2n} - Q_n \approx 16R_{2n} - R_{2n} = 15R_{2n}. \tag{64}$$

Therefore,

$$R_{2n} \approx \frac{Q_{2n} - Q_n}{15}. \tag{65}$$

Taking absolute values gives $|R_{2n}| \approx \frac{|Q_{2n} - Q_n|}{15}$. Since $R_{2n} = I - Q_{2n}$, the error of the finer approximation is estimated by

$$|I - Q_{2n}| \approx \frac{|Q_{2n} - Q_n|}{15}. \tag{66}$$

2.5. Example: integral of the function $\frac{\sin(x)}{x}$

Our final task is to use our implementation to approximate

$$\int_0^5 \frac{\sin(x)}{x}\, \mathrm{d}x. \tag{67}$$

At $x = 0$ it appears as if we are doing division by zero, but this is removable:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1. \tag{68}$$

The `sinc` function therefore defines `sinc(0) = 1` explicitly. This gives us a continuous and smooth function over the entire interval. Although the two Gauss–Legendre nodes lie strictly inside each subinterval, defining the value at zero correctly gives a complete description of the mathematical problem and also allows the function to be tested directly.

The sequence of approximations demonstrates fourth-order convergence, as seen in Table 2:

N	Evaluations	Q_N	Error estimate
16	32	1.549931574721095	$3.3264 \cdot 10^{-7}$
32	64	1.549931265514098	$2.0614 \cdot 10^{-8}$
64	128	1.549931246229614	$1.2856 \cdot 10^{-9}$
128	256	1.549931245024974	$8.0309 \cdot 10^{-11}$
256	512	1.549931244949693	$5.0187 \cdot 10^{-12}$

Table 2: Convergence of the composite two-point formula. The estimate in row N compares the approximations for $\frac{N}{2}$ and N.

For $N = 128$, we obtain

$$Q_{128} = 1.549931245024974, \quad \hat{E}_{128} = \frac{|Q_{128} - Q_{64}|}{15} = 8.0309 \cdot 10^{-11}, \tag{69}$$

so the estimated relative error is

$$\frac{\hat{E}_{128}}{|Q_{128}|} \approx 5.18 \cdot 10^{-11} < 10^{-10}. \tag{70}$$

Thus $N = 128$ is the first successful doubling step. The final rule has 128 subintervals and uses two nodes in each one, so it requires 256 function evaluations. The search process successively evaluated the rules for $N = 1, 2, 4, ..., 128$ and used 510 function evaluations without The full search performed 510 function evaluations, which describes the cost of the search, while the 256 is the cost of the final approximation.

The Richardson-extrapolated result is 1.549931244944664, while the non-extrapolated result of the prescribed rule is 1.549931245024974.

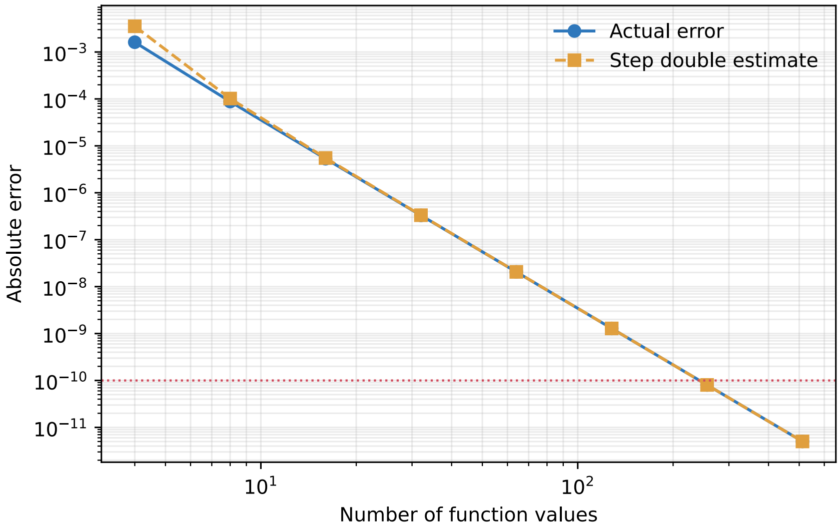

Figure 4: The actual error relative to the reference value and the estimate obtained by doubling. The red dotted line is the tolerance 10^{-10}.